# Yue Lin



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## Education

- **The Chinese University of Hong Kong, Shenzhen** — Shenzhen, China
  *PhD Student - School of Data Science* — 2024.8 - Present
  Research Interests: Multi-Agent Reinforcement Learning, Information Design.
  Fortunately advised by Prof. Baoxiang Wang and Prof. Hongyuan Zha.
  Other close collaborators include Prof. Wenhao Li and Prof. Pascal Poupart.

- **Tianjin Polytechnic University** — Tianjin, China
  *Bachelor of Engineering - Computer Science and Technology* — 2018.9 - 2022.6
  - **School of Computer Science and Technology** — 2019.9 - 2022.6
    GPA: 3.89/4 (92.2/100) Rank: 1/127
  - **School of Mechanical Engineering** — 2018.9 - 2019.6
    GPA: 3.90/4 (92.0/100) Rank: 1/60

## Experience

- **Microsoft** — China
  *Research Intern @ Microsoft Research Asia (MSRA)* — 2026.8 - Present
  Group: Data, Knowledge and Intelligence.
  Mentor: Lu Wang.

- **Tencent** — Shenzhen, China
  *Research Intern - Lightspeed Studios* — 2025.12 - 2026.6
  Group: Interactive Entertainment Group (IEG).
  Team: Omni.
  Mentor: Yuan Liu.

- **The Chinese University of Hong Kong, Shenzhen** — Shenzhen, China
  *Research Assistant - School of Data Science* — 2022.2 - 2024.8
  Fortunately advised by Prof. Baoxiang Wang and Prof. Hongyuan Zha.

## Publications & Preprints

- Information Design in Multi-Agent Reinforcement Learning.
  **Yue Lin**, Wenhao Li, Hongyuan Zha, Baoxiang Wang.
  *Advances in Neural Information Processing Systems (NeurIPS). 2023. Poster.*
  - This is one of my representative works.
  - As of 2026, my advisor also regards this as one of his representative works.
  - TLDR: Introduces a Markov signaling game and an obedience-constrained signaling-gradient algorithm that learns what information to reveal over time so self-interested receivers are willing to follow the sender's recommendations.

- Verbalized Bayesian Persuasion.
  Wenhao Li, **Yue Lin**, Yun Hua, Xiangfeng Wang, Bo Jin, Hongyuan Zha, Baoxiang Wang.
  *International Conference on Machine Learning (ICML). 2026. Poster.*
  - TLDR: Introduces a verbalized Bayesian-persuasion framework in which LLMs play sender and receiver while an equilibrium solver searches for natural-language messages the receiver has reason to follow, including in multi-stage dialogues.

- Policy-Conditioned Policies for Multi-Agent Task Solving.
  **Yue Lin**, Shuhui Zhu, Wenhao Li, Ang Li, Dan Qiao, Pascal Poupart, Hongyuan Zha, Baoxiang Wang.
  *arXiv preprint. 2025-12-24.*
  - This work was scooped by *Evaluating LLMs in Open-Source Games*. We then planned a substantially improved version, but that follow-up direction was also scooped by Google DeepMind's *Code-Space Response Oracles: Generating Interpretable Multi-Agent Policies with Large Language Models*.
  - TLDR: Represents each agent's strategy as readable source code and introduces Programmatic Iterated Best Response, where an LLM rewrites one policy against another using game rewards and unit tests as feedback.

- History-Dependent Policy Gradient.
  **Yue Lin**, Jiacheng Nie, Hongyuan Zha, Baoxiang Wang.
  *Draft. 2024.*
  - No paper was produced; our method was scooped by Google's paper *Multi-agent cooperation through learning-aware policy gradients* (*ICLR 2025*).
  - TLDR: Derives an exact policy gradient for non-stationary multi-agent learning systems by explicitly accounting for the histories of the other agents.
- The Reciprocity Gradient.
  **Yue Lin**, Pascal Poupart, Shuhui Zhu, Dan Qiao, Wenhao Li, Yuan Liu, Hongyuan Zha, Baoxiang Wang.
  *Advances in Neural Information Processing Systems (NeurIPS). 2026. Poster.*
  - This is one of my representative works. Review scores: 6/5/4 (out of 6; 4 = *borderline accept*).
  - TLDR: Derives a pure-RL reciprocity gradient that solves cross-agent influence attribution by differentiating future returns through chains of actions, evaluations, and other agents' responses, so an agent learns from scratch both how to behave under social scrutiny and how to reshape group norms—without reward shaping, centralized mechanisms, or reputation priors.
- Talk, Judge, Cooperate: Gossip-Driven Indirect Reciprocity in Self-Interested LLM Agents.
  Shuhui Zhu, **Yue Lin**, Shriya Kaistha, Wenhao Li, Baoxiang Wang, Hongyuan Zha, Gillian K. Hadfield, Pascal Poupart.
  *International Conference on Machine Learning (ICML). 2026. Poster.*
  - TLDR: Introduces ALIGN, a decentralized mechanism in which self-interested language-model agents exchange open-ended, tone-graded gossip to build reputations, coordinate social norms, and isolate defectors.
- Information Bargaining: Bilateral Commitment in Bayesian Persuasion.
  **Yue Lin**, Shuhui Zhu, William A Cunningham, Wenhao Li, Pascal Poupart, Hongyuan Zha, Baoxiang Wang.
  *Workshop on Information Economics × Large Language Models at the ACM Conference on Economics and Computation (EC). 2025. Poster.*
  - TLDR: Recasts long-term Bayesian persuasion as two-sided bargaining with commitment on both sides, separating the value of private information from the advantage of proposing first.
- Innovative Design and Simulation of a Transformable Robot with Flexibility and Versatility, RHex-T3.
  **Yue Lin**, Yujia Tian, Yongjiang Xue, Shujun Han, Huaiyu Zhang, Wenxin Lai, Xuan Xiao.
  *International Conference on Robotics and Automation (ICRA). 2021. Oral.*
  - Delivered a presentation at the Xi'an conference venue.
  - TLDR: Introduces a two-degree-of-freedom transformable RHex design that switches among wheel, leg, claw, and hook modes for efficient travel, obstacle crossing, object transport, and ladder climbing.
- A snake-inspired path planning algorithm based on reinforcement learning and self-motion for hyper-redundant manipulators.
  **Yue Lin**, Jianming Wang, Xuan Xiao, Ji Qu, Fatao Qin.
  *International Journal of Advanced Robotic Systems (IJARS). 2022.*
  - TLDR: Introduces Swinging Search and Crawling Control, which uses reinforcement learning over the arm's self-motion to find a collision-free target configuration, then moves the arm toward it by crawling instead of searching every intermediate pose.

## Patents

- Method and Apparatus for Pretraining Large Language Models Based on Federated Learning
  (基于联邦学习的大语言模型预训练方法和装置).
  Ang Li, Baoxiang Wang, Hongyuan Zha, **Yue Lin**.
  *Chinese Invention Patent.*
  No. ZL 2026 1 0720098.0 (Grant No. CN 122242656 B), Granted Jul. 2026.
- Communication Method, Terminal Device, and Storage Medium for Multi-Agent Reinforcement Learning
  (多智能体强化学习通信方法、终端设备及存储介质).
  **Yue Lin**, Wenhao Li, Hongyuan Zha, Baoxiang Wang.
  *Chinese Invention Patent.*
  No. ZL 2023 1 0397744.0 (Grant No. CN 116455754 B), Granted Sep. 2025.

## Professional Services

- NeurIPS 2025 Top Reviewer
- ICML 2026 Silver Reviewer
- Independent reviewer for top-tier venues:
  - NeurIPS 2024 [6; 45615]; 2025 [5; 32912]; 2026 [4; 17738]

- ICLR 2025 [3; 21831]; 2026 [1; 5994]; 2027 [5; 13765]
- ICML 2025 [6; 32893]; 2026 [6; 43185]
- AAAI 2027 [6; 20898]
- AAMAS 2027 [5; 0]
- TMLR 2025 [2; 38363]; 2026 [2; 10331]

- Volunteer:
  - ACL 2026 [1; 4155]
  - AAMAS 2024 [3; 9876]; 2025 [2; 4792]; 2026 [2; 2303]
  - ICML (Position) 2025 [2; 5016]

- Format: [assigned manuscripts; review characters]. First 0: invitation accepted, awaiting assignment; second 0: assigned, review pending. Total reviews: 61.
- Teaching Assistant @ The Chinese University of Hong Kong, Shenzhen:
  - [MFE5340] Artificial Intelligence in Financial Engineering: Quantitative Investment (2026-27 Term 1)
  - [CSC6021/AIR6001] Artificial Intelligence (2024-25 Term 2)